

A Study on the Design Scheme of a 2540 PPI Fast LCD for VR[†]

Yanqing Chen*, Xinfang Li*, Jinyao Wang*, Shibiao Wang*, Jie Tong*, Jianyun Xie*, Yunbo Zhang*

*BOE CHUANGYUAN Technology Co., Ltd., Beijing, China

Abstract

In this study, a feasible design scheme for a 2540-PPI Fast liquid crystal display (LCD) is introduced. To reduce process complexity and enhance panel transmittance, a Delta pixel structure and a novel color filter-on-array (COA) pixel design scheme is adopted. In addition, a Mini-LED backlighting is used to achieve a wide color gamut and high contrast ratio. This design realizes an ultra-high PPI display suitable for next-generation virtual reality (VR) products.

Author Keywords

2540 PPI; Fast LCD; Delta pixel structure; pixel design; Mini-LED backlight.

1. Introduction

With the continuous technological progress and iterative upgrades of virtual reality (VR) and mixed reality products, the demand for ultra-high-resolution displays for deep user immersion continues to increase [1-3]. At present, ultra-high-PPI products, such as those used in Apple's Vision Pro, predominantly utilize organic light-emitting diode on silicon (OLEDoS) displays [4,5]. However, high manufacturing costs and significant process difficulties increase the cost of the final product. Fast liquid crystal display (LCD)-based VR screens have lower costs; however, their PPI is considerably lower than that of OLEDoS displays, resulting in a considerable gap in visual quality [6,7]. Therefore, low-cost, ultra-high-PPI Fast LCDs must be manufactured.

Mainstream Fast LCD VR products currently achieve 1200–1500 PPI. Further enhancement in PPI is hindered primarily by process complexity and overall power consumption [8]. Pixel size decreases with increasing PPI; however, the dimensions of thin-film transistor (TFT) and process precision cannot be simultaneously optimized; this makes TFT routing layouts extremely difficult to achieve. High-PPI Fast LCDs require a color filter-on-array (COA) design structure [9,10], which imposes high demands on the morphology of RGB resist and considerably increases the overall process difficulty. In addition, the dimensions of key light-blocking layers cannot be reduced further while ensuring optimal visual performance, which reduces aperture ratio, decreases transmittance, and increases the overall power consumption.

To address these issues, a design scheme for a 2540-PPI Fast LCD is proposed herein. By adopting a Delta pixel structure and optimizing the pixel design, process complexity and overall power consumption are reduced. Furthermore, the gate-on-array (GOA) design is optimized to further minimize the bezel size. In addition, Mini-LED backlighting is employed to enhance image resolution, achieving a wide color gamut and high contrast ratio.

2. Delta Pixel Structure

In conventional real RGB arrangements commonly used in display panels, each pixel contains three subpixels; these physically constrain resolution enhancement. As shown in Figure 1, this real RGB layout exhibits lower fineness compared with the RGB Delta arrangement within the same area. Due to

its regular rectangular grid structure, it tends to produce noticeable “staircase” aliasing when displaying diagonal lines.

In contrast, the triangular layout of the RGB Delta arrangement breaks the conventional rectangular grid pattern; this results in more natural transitions and smoother rendering of curves and diagonal lines, effectively suppressing aliasing artifacts. In addition, with the same number of physical subpixels, the integration of sophisticated subpixel rendering algorithms enables the rendering of additional “pixel points.” Consequently, a higher apparent PPI and a perceptually finer image are obtained, which are particularly critical for VR/AR products that demand high PPI.

The RGB Delta pixel layout increases the horizontal subpixel pitch to 1.5 times that of the real RGB arrangement, offering enhanced compatibility with TFT fabrication processes and enhancing panel reliability.

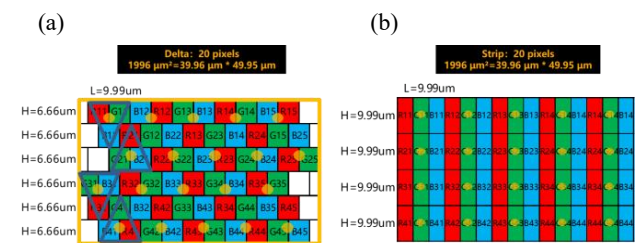


Figure 1. (a) RGB Delta and (b) RGB real pixel arrangement

3. COA Pixel Design

3.1 Self-Aligned Photo Spacer Design

In high-PPI LCD panels, photo spacer (PS) design is a critical factor that influences transmittance. This is because PS structures can disrupt liquid crystal (LC) alignment, causing light leakage in the dark state. A shielding layer (SL) is typically placed on the color filter (CF) substrate to block such leakage. The shielding distance must cover the abnormal LC region around the PS and account for overlay variation between the SL and PS and the variations in their critical dimensions (CDs). This requirement considerably reduces the aperture ratio.

To address this limitation, we propose a horizontal photo spacer1 (PS1) design. Unlike the conventional process of patterning the SL first and then forming the PS, this design approach involves coating the first metal shielding layer (SL1) without exposure and etching, directly applying the PS material, and simultaneously etching both the PS and SL1. This eliminates overlay error between PS1 and SL1 and reduces the required shielding margin, as shown in Figure 2. Using a new PS material, the CD of PS1 can be reduced to 1 μm and the SL1 width can be reduced; this will ultimately increase the aperture ratio (AR) and improve panel transmittance. With these implementations, the AR is expected to increase by 8%.

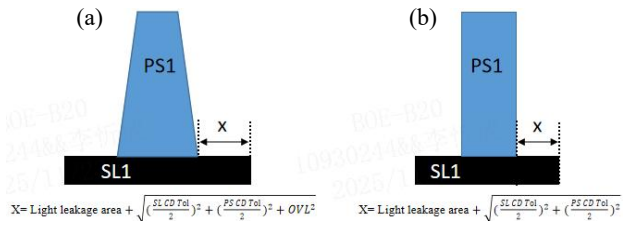


Figure 2. (a) Conventional PS design and (b) self-aligning PS design

3.2 Main SL Design

Figure 3 shows the main shielding layer2 (SL2) and shielding layer3 (SL3) on the array substrate. SL2 is arranged along the data lines and is formed using a low-reflectivity metal. Compared with a traditional strip or mesh-shaped SL2, this design uses island-shaped SL2 that can improve the LC efficiency by 10%. The thickness of the SL2 is increased to 0.4 μm , and at PS locations, the upper and lower parts are connected to serve as the photo spacer2 (PS2) on the array substrate (Figure 4). The unconnected island SL2 are considered to have minimal impact on the brightness of the black screen because they are parallel to the initial LC alignment direction. Although the connected SL2 may induce distortions in the LC display and cause black-state light leakage, SL3 can block these leakage points. This PS2 design scheme achieves a 15% higher AR compared with the traditional mesh-shaped PS design.

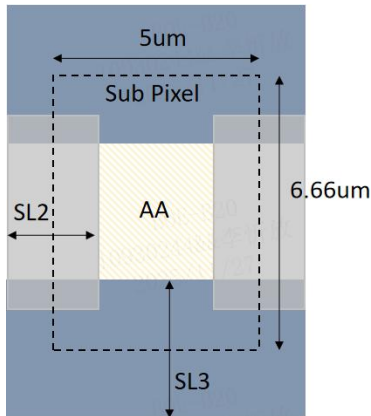


Figure 3. Design of shielding layers (SL2 and SL3)

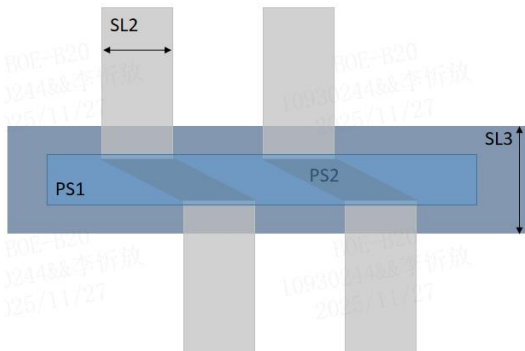


Figure 4. Design of PS1 and PS2

3.3 PLN0 Design

Compared with traditional LCDs that have separate TFT and CF substrates, the COA structure offers advantages, i.e., improved aperture ratio and reduced color shift. However, the COA

structure introduces process challenges. First, slots must be formed between adjacent color resists in the Y-direction to ensure electrical connectivity. Second, to maintain color gamut, these slots must remain within the shielding region after accounting for process variation; this restriction imposes strict requirements on color-resist patterning.

In a 2540-PPI design, the vertical subpixel pitch is only 6.66 μm and the SL width is as small as 3.32 μm . Thus, the color-resist slot must fit within this 3.32- μm range. The taper angle of existing color-resist materials is largely determined by their properties and cannot be easily adjust via process optimization; therefore, simultaneously meeting electrical connectivity and color gamut requirements at 2540 PPI becomes infeasible.

To overcome this limitation, we propose using PLN0 as a barrier between color resists. This approach eliminates limitations imposed by the color-resist taper angle. As shown in Figure 5, the dimension of PLN0 (b) is determined by the shielding layer width (a), which in turn defines the dimension of the color resist (c). This structure ensures electrical connectivity while achieving an sRGB color-gamut coverage of 100%.

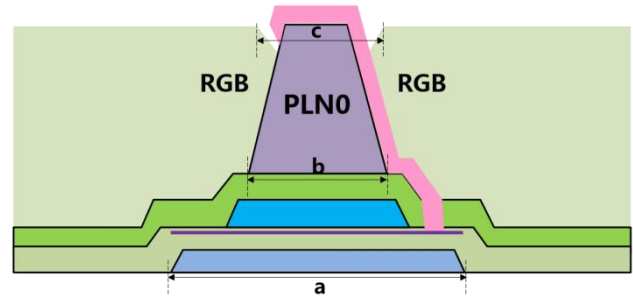


Figure 5. Schematic of PLN0

Furthermore, using a black organic material for PLN0 enables it to function as a low-reflection film in the horizontal direction. Compared with a conventional black matrix (BM) material, the overlay variation between PLN0 and SL3 is minimal because these layers are placed on the array substrate. Consequently, it achieves enhanced transmittance uniformity and brightness roll-off (BRO) performance compared with when BM is used as the low-reflection layer. The BRO performance can be particularly increased by almost 16%.

3.4 Indium Tin Oxide (ITO) Design

Figure 6 shows the previous ITO design, with a parallelogram-shaped slit. This design can create dark areas at the upper-left and lower-right corners, reducing LC efficiency. To eliminate these dark areas, the ITO slit was designed to be continuous through the area (Figure 7). This enabled normal LC rotation at the upper-left and lower-right corners, thereby improving LC efficiency. This design modification can increase the transmittance by 10%.

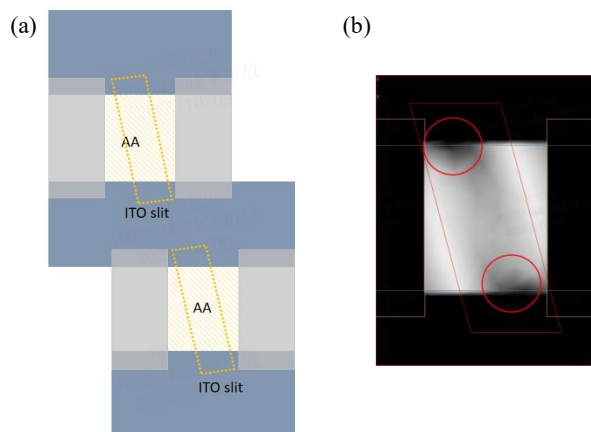


Figure 6. (a) Traditional ITO design and (b) LC efficiency diagrams

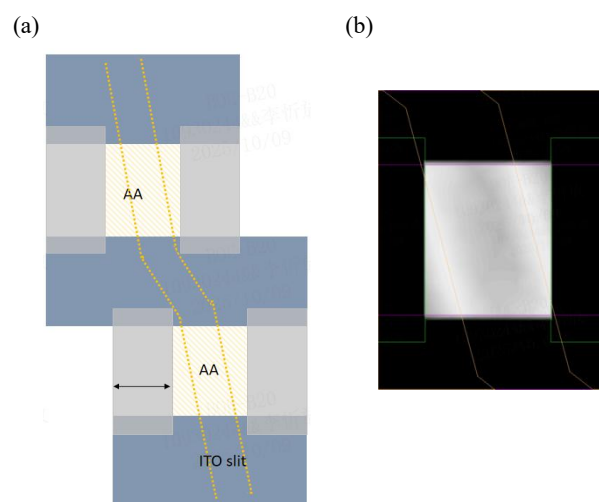


Figure 7. (a) New ITO design and (b) LC efficiency diagrams

4. GOA Design

As PPI increases and pixel size decreases, the vertical space available for GOA units is reduced. Single- or dual-gate-driven GOA circuits occupy more lateral space compared with multi-gate-driven GOA architectures, resulting in wider borders. Moreover, single- and dual-gate-driven designs require a large number of GOA units, leading to increased clock signal loading and switching power consumption. In contrast, a multigate-driven GOA circuit allows multiple pixel rows to share one GOA unit, thereby reducing both border width and panel power consumption.

The GOA model enables one GOA unit to drive four pixel rows, thereby enabling the product border to be reduced to 1 mm. Figure 8 shows the circuit diagram of the GOA model comprising 20 TFTs and 5 capacitors and is driven by 16 clock signals. The cascading relationship is illustrated in Figure 9. By adjusting the timing sequence, this GOA model can support eye-tracking functionality.

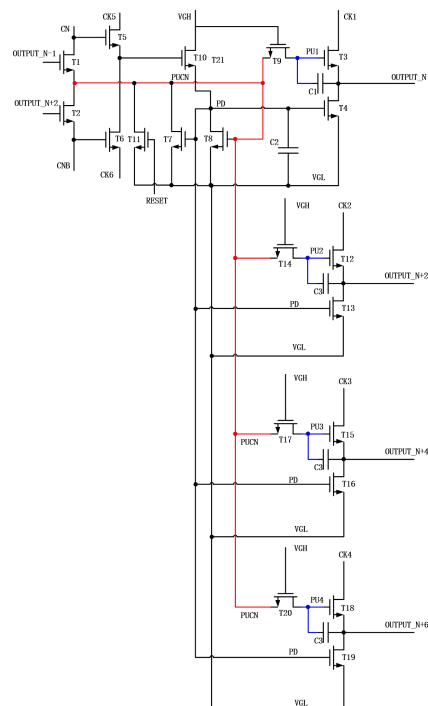


Figure 8. Circuit diagram of the GOA model

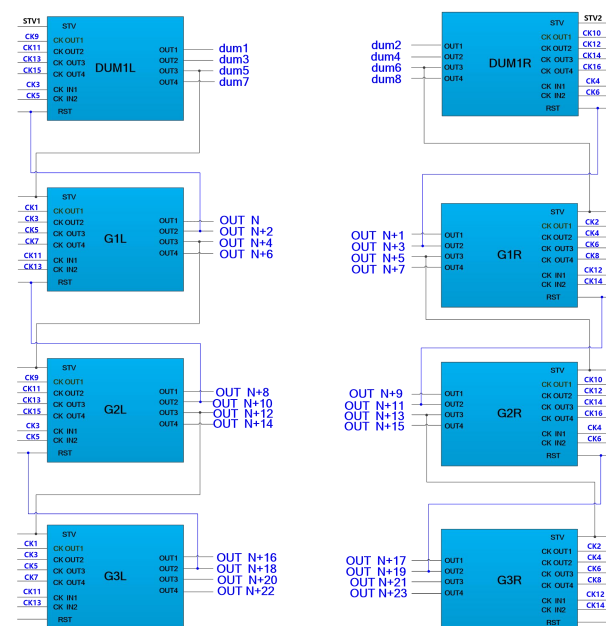


Figure 9. Schematic of the cascading relationship

5. BLU Design

The 2540-PPI module has a display resolution of 3478×5582 . To ensure smooth display without any lag, a refresh rate of at least 90 Hz is required. At 90 Hz, the frame time is 11.11 ms and the row scan time is 1.5 microseconds; the total scan time is 8.37 ms. An edge-lit backlight with black-frame insertion requires a duty time of 1.11 ms, and the LCD response time must be 1.63 ms. However, the current response time only reaches 3.6 ms, rendering the edge-lit backlighting with black-frame insertion unsuitable; this process is shown in Figure 10.

Therefore, this design adopts a direct-lit backlight solution because it offers the following advantages: 1. capability for high refresh rates; 2. ability to achieve a wider color gamut by incorporating quantum dot (QD) films; 3. Ability to achieve a high contrast ratio via the multizone local dimming of the backlight, coupled with local dimming algorithms.

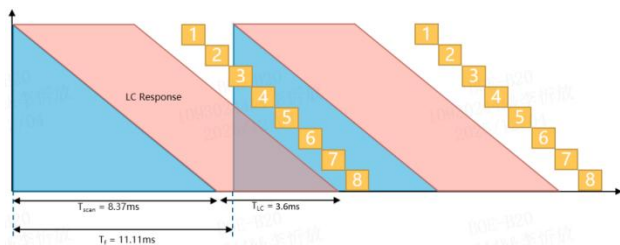


Figure 10. Edge-lit backlight with a black plug lit up

In the direct-lit backlight scheme, LEDs in the light board part are arranged in a 48 (rows) $\times$ 45 (columns) configuration, totaling 1950 LEDs. The design uses five LEDs within one zone, resulting in 390 zones. We adopt the 0407 LED chip. By designing a specific white rubber barrier and transparent rubber filling on the light board, the single-zone halo PSF is narrowed to 10 mm (full width corresponding to 1% Lv). This single-zone PSF narrowing considerably enhances the contrast of the displayed image, as shown in Figure 11. The film-stack architecture is shown in Figure 12(a), indicating the following contents from bottom to top: light board, transparent blue/reflective red-green film, QD film, bottom diffuser, top diffuser, bottom prism, top prism, and LCD panel. Figure 12(b) shows the backlight structure, where the LED driver module is placed inside the backlight.

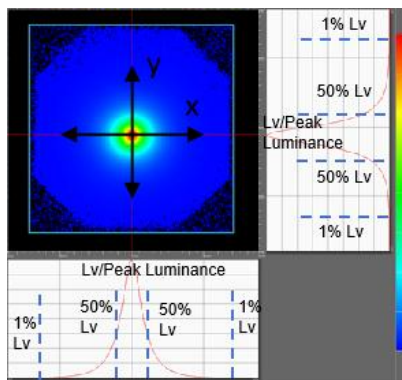


Figure 11. Single-zone halo PSF

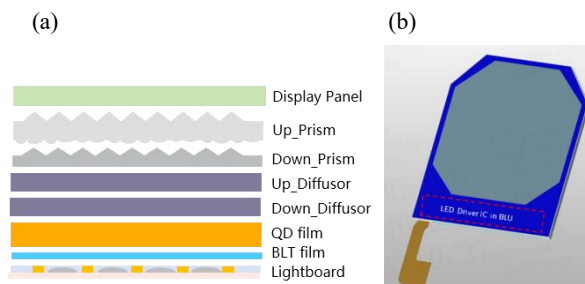


Figure 12. The edge-lit backlight with a black plug lit up.

6. Conclusion and Significance

Herein, a feasible design for a 2540-PPI Fast LCD was

presented. The TFT process complexity was reduced by adopting a Delta pixel structure. The introduction of the PLN0 scheme enhanced the pixel design. In addition, the PS structure and material properties were improved, and the SL2 and ITO designs were optimized. These improvements ultimately enhanced the panel transmittance and reduced the overall power consumption. The GOA layout was also optimized to achieve a narrow bezel. In addition, image resolution, color gamut, and contrast ratio were improved using a Mini-LED backlight. Thus, the proposed design offers a competitive solution for next-generation VR displays.

7. References

- Rao, L., Park, Y., Klement, A., Kang, C., Park, E., Zhuang, J., ... & Shen, S. (2023, June). 5-1: Invited Paper: Infinite Display for Meta Quest Pro. In *SID Symposium Digest of Technical Papers* (Vol. 54, No. 1, pp.32-35).
- Wu, Y. H., Tsai, C. H., Wu, Y. H., Cherng, Y. S., Tai, M. J., Huang, P., ... & Yang, C. L. (2023). Breaking the limits of virtual reality display resolution: the advancements of a 2117-pixels per inch 4K virtual reality liquid crystal display. *Journal of Optical Microsystems*, 3(4), 041208- 041208.
- Wu, Y. H., Chang, C. C., Tsou, Y. S., Lo, Y. C., Tsai, C. H., Lu, C. H., ... & Chuang, F. M. (2023). Enhancing virtual reality with high-resolution light field liquid crystal display technology. *Journal of Optical Microsystems*, 3(4), 041202- 041202.
- Waisberg, E., Ong, J., Masalkhi, M., Zaman, N., Sarker, P., Lee, A. G., & Tavakkoli, A. (2024). The future of ophthalmology and vision science with the Apple Vision Pro. *Eye*, 38(2), 242-243.
- Kaye, A. D., Islam, R. K., Islam, K. N., Khachemoune, A., Haas, C., Barrie, S., ... & Islam, R. (2024). Apple vision pro and its implications in mols micrographic surgery: A narrative review. *Cureus*, 16(10).
- Sim, I., Choi, K., Baek, Y., Choi, J. H., Jo, J., Yeom, J., ... & Lee, K. (2025). Microdisplay technologies in augmented reality and virtual reality headsets. *Nature Reviews Electrical Engineering*, 2(9), 634-650.
- Jung, D., Cho, A., Kim, H., Jeong, M., & Lee, J. H. (2023, June). 32 - 4: Late - News Paper: A 1.03 - inch 2560x2560 3514 PPI Low Power OLEDs Backplane with In - pixel Up - scaling Technique. In *SID Symposium Digest of Technical Papers* (Vol. 54, No. 1, pp. 465-468).
- Zhang, Z., Zhang, Z. Y., Xuan, M. H., Wang, L. Z., Tong, B. B., DI, Y. P., ... & Yu, L. F. (2023, June). 5 - 5: Development of 1500PPI VR LCD display device based on LTPO process structure. In *SID Symposium Digest of Technical Papers* (Vol. 54, No. 1, pp. 47-49).
- Watakabe, H., Jinnai, T., Suzumura, I., Hanada, A., Onodera, R., Tada, M., Mochizuki, K., Tanaka, H., & Ito, T. (2019). Development of advanced LTPS TFT technology for low power consumption and narrow border LCDs. *SID Symposium Digest of Technical Papers*, 48(1), 541-544.
- Chang, T. K., Lin, C. W., & Chang, S. (2019, June). 39-3: invited paper: LTPO TFT technology for AMOLEDs. In *SID Symposium Digest of technical papers* (Vol. 50, No. 1, pp. 545-548).